Audit, step 2: independent review of the brainstorm

Reviewer: a fresh agent (model: Fable), read-only, 2026-10-20 (night). It did not write the code. Below is its report in full, followed by the actions taken in response.

Verified correct by the reviewer (no finding)

- **Matching derivation:** $Z_{in} = X_m^2 / (Z_c - jX_m)$, so $X = \sqrt{Z_0 R}$, $C_m = 1/(\omega X)$, $L_m = X/\omega$.
- **Network maths:** `s2z` ; `reduce()` (Schur complement); the shunt-C / series-L network in `external()` .
- **Quasi-static maths:** the AGM elliptic integrals and Maxwell's formula; the $\mu_0/4\pi$ unit constants.
- **openEMS interfaces:**
  - the phantom HDF5 layout matches openEMS's own writer;
  - the incident wave `uf_inc = 0.5(U + I·Z_ref)` matches the co-simulation's $a = (V + Z_0 I)/2$;
  - port direction and wire order are consistent;
  - the feed is the same break in `K.breaks` and `cut_wire` .
- **Thin-wire correction:** sign and a_eff inversion are arithmetically consistent (a_eff 0.17 mm from L_raw 278.6 nH).

Missed failure modes (M1–M17)

ID	Failure mode	Why it matters	Detect / test
M1	V15–V20 are not in the test suite; the app banner and `validation_status.js` say "all pass" while **V15 failed**	False validation status; integrity claim untrue	Aggregate `sim/out/*/cosim.json` into the report
M2	Stair-stepped `AddCurve` strands can short at crossings (1.5 mm apart on a 1 mm grid)	Knot topology destroyed in FDTD; V15b would fail with the wrong diagnosis	Trefoil at 0.5 vs 1.0 mm; crossing connectivity
M3	Frequency-domain dump vs port-probe normalisation across runs	Every B1 superposition (V18)	Compare a time-domain dump transformed in Python with the FD dump, or a superposed field with a direct run
M4	V19's d_0 is the *flat, unmirrored* null; the case is wrapped and mirrored	V19 tests a different configuration than registered	Report on-helmet k_qs; bracket the on-helmet null
M5	No air pair case; the sign of M_eff vs M_qs is never checked	Tissue, wrapping and capacitor effects cannot be separated	Add `pair_air_3T_d+0`; check the signs
M6	V17 evaluated with neighbours on 50 Ω; tuning used preamp terminations	V17 can fail for reasons unrelated to tuning	Report S11 under both terminations
M7	V18 normalises to the in-plane maximum (the noisiest point)	One voxel sets the comparison	Report a robust normalisation alongside
M8	V10 and V12 are self-reported (circular)	A reporting bug passes its own test	Recompute independently in the test
M9	V14 compares thinned wires with thinned wires	Thinning error untested	Compare thinned vs full-wire field
M10	Calibration file untracked, keyed only by grid; sim outputs carry no git hash; outputs came from uncommitted code	Not reproducible	Record hash and options; track the calibration
M11	`dist/` is stale	The shared package runs older code	Rebuild; check builder consistency
M12	Design §6b says `AddWire`; the code uses `AddCurve`	Documentation and code disagree	Update the document
M13	The coronal plane is "15 mm inside the head" measured from the helmet, so only about 5 mm into the skin	V18 near-surface points are the worst stair-stepped ones	Define the plane from the head surface
M14	The phantom's neck cut is ignored in the depth calculation	Points misclassified near zMin	depth = min(ellipsoid depth, z − zMin)

ID	Failure mode	Why it matters	Detect / test
M15	In air, Re(Z) of a PEC loop is at noise level	Air C_m and L_m reported with false precision	Flag when R_loop is tiny
M16	`sources/` PDFs and the GPL imports in a package meant for sharing; NOTICE lists Three.js only	Licence and copyright exposure	Licence inventory
M17	`tests/run.js` header says “V1–V6, git 39a4e83”	Readers can’t trace registrations	Add a per-check `registered_in`

Mis-rated brainstorm items

Item	Reviewer’s correction
C5	Should be *Partial* (rests on the solver’s own self-report)
C3	V9 has limited coverage: no cap, odd-N or midline-heavy case
B12	The singular branch is untested
C12	Low risk
D1	Understated: V15b is non-circular only through its transfer to the trefoil
A14	V18 compares a circularly polarised component with a linear magnitude; this is a physics difference the check will count against the quasi-static model
E4	Should be *Open*
B8	The `z.imag > 0` filter is vacuous

Reviewer’s top 10

1. M1
2. M2
3. M3
4. A7 / V15
5. M4 / M5
6. A9 / A10
7. M10 / C14
8. M8
9. M6 / M7
10. M11 / M12

Actions (filled in as they are done)

See DECISIONS.md, “Audit actions”.